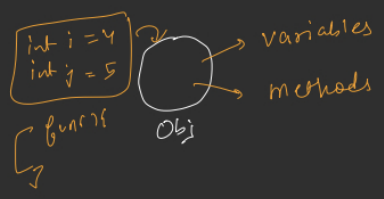
Immutable class




```
Class Student {
    int age;
    String name;
    Student (int a, String n){
        =
    }
    // getters
    // setters
    markAttendance() {-}
}
```

```
Student s1 = new Student
        (26, "Aditya");

s1.setName("Rohit");

Class CompSciStudent
        extends Student {
    @override
    void markAttendance() {
        =
    }
}
```

Rules of Immutable Object

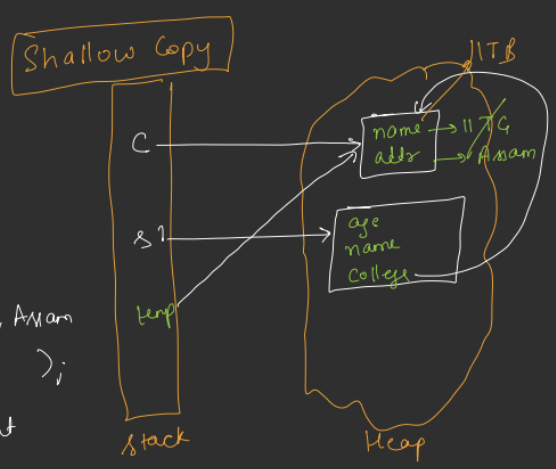
- Mark my class as final.
- Mark my variables as private & final.
- No setters

```
Class College {
    String name
    String address;
}

Class Student {
    College college;
}

College c = new College("IITG, Assam");
Student s1 = new Student
        (26, Aditya, c);

s1.getCollege().name = IITB
```



↳ Defensive copy in Constructor / getter.

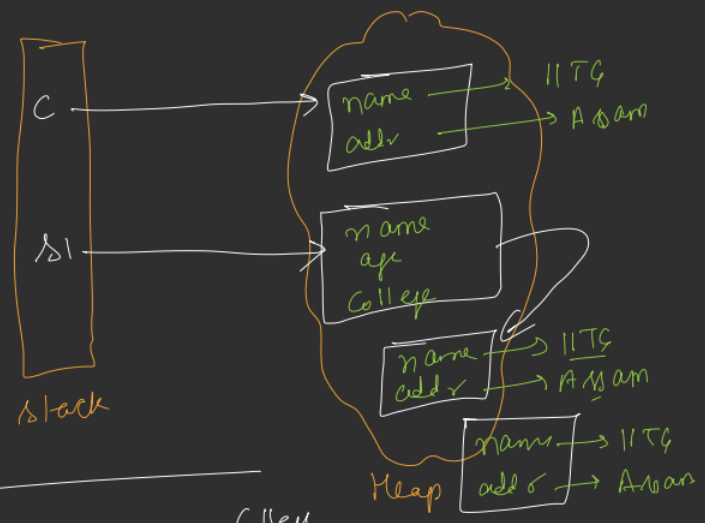
↳ Defensive copy in Constructors / getters.

```
final Class Student {
    private final name;
    private final age;
    private final College;

    Student (name, age, college) {
        this.college = new
                College (college.name,
                         college.addr);
    }

    public College getCollege {
        return new College(
                this.college.name,
                this.college.addr);
    }
}
```

Deep Copy




```
College c = new College
            (IITG, Assam);

Student s1 = new Student(
            28, Aditya c);

s1.getCollege();
```

Threads

↳ Race Condition